Sexism Detection in Tweets

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Abstract

This study investigates the prevalence and detection of sexism in English and Spanish tweets, leveraging Natural Language Processing (NLP) techniques. Based on the EXIST 2024, 2023, and 2021 Sexism Identification in Social Networks and Memes dataset, which includes annotated social media content for the identification of sexist language in both text and meme contexts, our research aims to advance the understanding of sexist expressions on digital platforms. We employ a range of methodologies, from classical machine learning models like Logistic Regression, Naive Bayes, and SVM to state-of-the-art NLP techniques, such as contextualized word embeddings and deep learning models, to capture subtle linguistic patterns associated with sexist discourse. Our results demonstrate the efficiency of the transformers based models but also the Logistic Regression in detecting sexist behavior, contributing to automated content moderation systems aimed at promoting gender equality in online environments.

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1 Introduction

Sexism in social media platforms like Twitter has become a pervasive issue, with harmful gender biases influencing online discourse. These biases range from overt hate speech to subtle forms of discrimination. As social media becomes an integral part of public conversations, understanding and mitigating harmful content is essential. However, detecting sexism in online platforms is challenging due to varied linguistic expressions, implicit biases, and contextual nuances that complicate the identification process.

Recent advancements in Natural Language Processing (NLP) have significantly contributed to the development of automated systems for detecting harmful content, such as hate speech and misogyny. However, detecting sexism, which can appear in diverse and nuanced forms, remains underexplored, particularly in the context of Twitter and other rapidly evolving platforms. Moreover, while existing datasets for detecting hate speech and misogyny are available, many are limited by language scope or lack comprehensive data.

This study addresses these gaps by utilizing the EXIST 2024, 2023, and 2021 Sexism Identification in Social Networks and Memes datasets, which provide annotated examples of social media posts and memes spanning various forms of sexism. We leverage these datasets to explore the intersection of language,image and online behaviors, aiming to create a robust methodology for identifying both explicit and implicit sexism in tweets.

The approach to this study is divided into two distinct parts to effectively address the unique challenges posed by each component of the dataset. In the first part, we will focus on the textual analysis of the dataset, where the primary objective is to explore the relationship between the text content and the presence of sexism. This section involves preprocessing the textual data, tokenizing the content, and employing natural language processing (NLP) models to classify whether the text in each tweet is sexist or non-sexist. Our methodology will involve the use of state-of-the-art transformer models, such as BERT or XLM-RoBERTa, to capture semantic nuances and context in the text, ensuring that the model can differentiate between sexist and non-sexist statements. In the second part, we will shift our attention to analyzing memes, which present an additional layer of complexity due to their multimodal nature. Memes often combine both text and images, making them harder to classify using text-based models alone. In this section, we will focus on handling both the textual and visual components of the memes. Since the text is already provided separately, the challenge will be in exploring how the combination of textual content and visual elements influences the classification of memes as sexist or non-sexist. We will adapt our models to handle this multimodal data by incorporating computer vision techniques alongside text-based transformers to gain a holistic understanding of the memes. The second part will therefore explore how both visual cues and text interact to determine the final classification of a meme.

2 Related Work

Sexism, defined as prejudice or discrimination based on gender, is a prevalent issue that is amplified by online platforms. Many researchers have made progress in developing automated systems for detecting sexism in social media content. These systems use a range of techniques, from rule-based methods to more advanced machine learning models.

There have been several approaches to classify hate speech spreaders online. Bag-of-words (BOW) techniques were the foundation for the first efforts on hate speech identification [4, 11, 12]. One of the earliest studies, published in 2012, used machine learning-based classifiers [13] rather than pattern-based techniques [8] for detecting abusive language. To identify hate speech, many traditional machine learning techniques have been used, including decision trees [4], logistic regression [11], and support vector machines. Recently Das et al. [3] used BERT-TFIDF based approach for profiling irony and stereotype spreading authors. There are various methods that use specific types of sentiment data as features.

The first edition of the EXIST challenge highlighted several promising methods for identifying sexism in social media content. For instance, [9] proposed a system leveraging both multilingual and monolingual BERT models, with a data translation and ensemble strategy for detecting and classifying sexism in English and Spanish. Similarly, [6] applied a multi-task learning approach to handle different tasks simultaneously, improving performance by leveraging shared information across tasks. Another significant approach was introduced by [10], who combined the hidden states of XLM-RoBERTa with a TextCNN, significantly improving performance by incorporating abusive word lexicons alongside the transformer model.

The second-place team in the EXIST2022 challenge [1] employed an ensemble of five models for Spanish and five models for English, including XLM-R, RoBERTa, BERT, and ALBERT. They also translated English tweets to Spanish and used token masking to augment their data. Meanwhile, the third-place team [7] demonstrated the utility of ensemble models combined with linguistic features, showing that these techniques could improve performance. The team that won the EXIST2023 challenge [5] focused on transformers, particularly mBERT and XLM-RoBERTa, solidifying their dominance in this task.

Our work contextualizes within the state of the art by using both Binary Relevance and Classifier Chain architectures alongside the XLM-RoBERTa model family, with the aim of balancing computational efficiency with robust performance. Additionally, several methods discussed by [2] for french corpus influenced the methodologies employed in our research.

3 Datasets

We fine-tuned our models using the training dataset from the EXIST shared task. The final dataset is a merge of the EXIST 2021, 2023, and 2024 datasets, which together contain a collection of 14,935 tweets.

The dataset includes tweets from two languages: Spanish and English, collected from two platforms—Twitter and Gab Social. Each tweet is annotated with information on the presence of sexism and its categorization if sexism is present. The tweets that do not contain sexism are labeled as non-sexist, while tweets with sexism are labeled as sexist, and classified into one of five categories: ideological-inequality, stereotyping-dominance, objectification, sexual-violence, or misogyny-non-sexual-violence. For this study, we focus on the first task—detecting sexism regardless of the category. The dataset shows that 51.9% of the samples are in Spanish and 48.1% are in English. The first task shows a relatively balanced distribution between sexist and non-sexist tweets. However, the second task exhibits a highly imbalanced distribution, with a disproportionate number of non-sexist samples. Among the sexist categories, ideological-inequality is the most represented, while objectification and sexual-violence have fewer examples.

3.1 Dataset Splitting

To guarantee an even distribution of memes in Spanish and English, the dataset is carefully divided into training and test sets. In order to maintain the representativeness of the training data and ensure that the models trained on it can effectively generalise to new examples across other languages, stratified splitting is essential. The test set acts as an impartial set to assess how well the machine learning models perform; the training set is utilised to fit the models. This process is necessary to determine how effectively the models will function in practical situations and to adjust hyperparameters to avoid overfitting. Additionally, by preventing the models from becoming biased in favour of any one language, the balanced distribution improves the models cross-linguistic applicability.

Language	**Train**	**Test**	**Sexist (%)**	**Not Sexist (%)**
English	5748	1437	52.7%	47.3%
Spanish	6200	1550	47.3%	52.7 %

Table 1: Data Sizes and Sexist/Not Sexist Distribution for English and Spanish

3.2 Preprocessing

Our preprocessing steps aimed to clean and enhance the raw dataset, which is inherently messy. we can use regular expressions to search through the tweets to see if they contained similar patterns. The pipeline utilizes Python's regex to remove user mentions, while preserving the meaningful content. Multilingual text processing is also included, with special attention to Spanish punctuation and spacing corrections. Additionally, I implemented text normalization techniques to compress repeated punctuation and characters, reduce excessive whitespace, and eliminate problematic symbols. This pipeline is optimized for production, serving as a foundational step in building robust NLP applications and machine learning models for social media analysis.

Approximately 34% of the tweets contain URLs or emojis, which are crucial for amplifying messages in such short texts. Instead of removing or replacing these URLs with tokens, we propose substituting them with the title found at the corresponding URL, which is time consuming (approximatively 1 hour for the whole dataset). Emojis were replaced by their detailed descriptions to capture their meaning more accurately, following the strategy proposed by Singh et al. (2019).

The word clouds for non-sexist and sexist tweets in both English and Spanish highlight the stark contrast in the language used. In the English word cloud for non-sexist tweets, the most frequent terms include neutral and positive words such as "forward," "week," "looking," and "games," suggesting discussions centered around everyday life and personal experiences without any harmful bias. In contrast, the word cloud for sexist tweets displays more aggressive and derogatory language, with words like "dont," "blame," "bastard," and "FFS," reflecting frustration, insult, and negative connotations often associated with sexism. Similarly, in the Spanish word clouds, non-sexist tweets contain terms like "GAMERGATE," "comicsgate," and "cambiado," indicating neutral discussions related to the gaming community and social events. On the other hand, sexist tweets feature offensive terms such as "madre" (mother), "capullo" (idiot), and "gringa" (a derogatory term for foreigners), demonstrating the use of gendered insults and toxic language aimed at women or marginalized groups. Both sets of word clouds underscore the presence of toxic and harmful discourse, especially in online communities related to gaming and social issues.

Figure 1: English Wordcloud

Figure 2: Spanish Wordcloud

4 Baselines Models

In this study, I employed a variety of **machine learning** techniques to detect sexism in tweets for each language separately, leveraging the unique linguistic properties of both English and Spanish. The workflow began with extensive **text cleaning** to preprocess the raw tweet data, followed by **lemmatization** to standardize words and make them consistent across the dataset. For this purpose, we used **spaCy's lemmatizer**, specifically the `en_core_web_sm` model for English and the `es_core_web_sm` model for Spanish. This lemmatizer integrates **rule-based** and **lookup-based** approaches, effectively reducing words to their base or dictionary form while preserving their semantic meaning. Unlike more basic **stemming** techniques, which can lead to overly crude reductions, spaCy's lemmatization accounts for **contextual nuances**. It uses part-of-speech tagging to distinguish between different meanings of the same word, resolving ambiguities like "walking" being reduced to "walk" when used as a verb, but leaving "walking" unchanged when used as a noun. The lemmatizer also handles irregular word forms through curated **exception tables**, converting words like "ran" to "run" and "geese" to "goose." This powerful linguistic tool achieves over **95% accuracy** on standard benchmarks while maintaining a minimal memory footprint (only **12MB** for the small model). Integrated seamlessly into **spaCy's** pipeline, it makes lemmatized forms available through the `token.__lemma_` attribute, without requiring any additional configuration, making it particularly effective for processing the informal language typically found in social media data, which often includes **slang**, **abbreviations**, and **contractions**.

Following lemmatization, we employed the **TF-IDF (Term Frequency-Inverse Document Frequency)** vectorization method to transform the cleaned text data into numerical features. This transformation helps capture the importance of terms relative to their frequency across the entire corpus, ensuring that distinctive and relevant words are prioritized. We set up a TF-IDF **vectorizer**, which generated training matrices of dimensions

(5748, 14280) for English and **(6200, 18323)** for Spanish, where the rows represent individual tweets and the columns correspond to unique terms extracted from the dataset.

To fine-tune the model, we employed **cross-validation** and **grid search** techniques, leveraging **Optuna** for hyperparameter optimization. The primary evaluation metric used during this tuning process was the **AUC (Area Under the Curve)** score, ensuring that the model was optimized for the best possible performance in distinguishing between sexist and non-sexist content. This combination of preprocessing, feature extraction, and hyperparameter tuning allows for effective and efficient model training, helping us achieve the best performance in detecting sexism across both English and Spanish tweet datasets.

For all the models, the English version consistently achieves slightly better scores compared to the Spanish version. This trend is observed across multiple evaluation metrics, including Accuracy and F1 Score. One possible explanation for this discrepancy is the use of a lemmatizer during the preprocessing stage for English text, which likely helped in normalizing the words and improving

Model	Accuracy	F1 Score
Logistic Regression (English)	0.7509	0.7421
Logistic Regression (Spanish)	0.7039	0.7067
Naive Bayes (English)	0.7279	0.7245
Naive Bayes (Spanish)	0.5335	0.6917
XGBoost (English)	0.7196	0.6883
XGBoost (Spanish)	0.6781	0.6745
SVM (English)	0.5595	0.1660
SVM (Spanish)	0.5271	0.6903

Table 2: Model Evaluation Results

the model's performance. On the other hand, the Spanish models may not have benefited from the same level of preprocessing, leading to lower scores overall. This observation suggests that the choice of preprocessing techniques, such as lemmatization, plays an important role in text classification tasks, and further experimentation with Spanish-specific preprocessing could improve the model's performance in that language.

While more sophisticated, methods like XGBoost and SVM require more computational resources and training time, Logistic regression and Naive Bayes are computationally more efficient and faster to train, making them more practical for situations where resources are limited. The results show that Logistic regression and Naive Bayes outperform them in this specific case due to their simplicity, better fit for the dataset's characteristics, and efficiency in training.

5 Tansformers Based Models

5.1 BERT

For a brief overview, BERT is a pre-trained language model that utilizes a deep bidirectional transformer architecture to generate context-sensitive representations of words. BERT is bidirectional in the sense that it processes text by considering both the left and right context of a word simultaneously, making it better at capturing context compared to models that only consider one direction. It uses a transformer architecture, which relies on attention mechanisms to process entire sentences or documents in parallel, as opposed to older architectures like recurrent neural networks (RNNs), which process data sequentially. We employ two BERT models (one model for each language) paired with a BERT Byte Pair Encoding (BPE) tokenizer for each language, enabling efficient tokenization and understanding of text. The BERT models are pre-trained on large text corpora and fine-tuned for specific downstream tasks. The BPE tokenizer is specifically chosen because it is effective at handling rare and out-of-vocabulary words by breaking them into subword units, ensuring that the tokenizer performs well even on complex and highly diverse texts.

Metric	Spanish Model	English Model	Overall
Precision (Non-Sexist)	0.76	0.85	
Precision (Sexist)	0.76	0.75	
Recall (Non-Sexist)	0.72	0.75	
Recall (Sexist)	0.79	0.85	
F1-Score (Non-Sexist)	0.74	0.80	
F1-Score (Sexist)	0.78	0.80	
Accuracy	0.76	0.80	**0.777**
F1-Score (Macro Average)	0.78	0.80	**0.787**

Table 3: Model Performance for Bert

5.2 XLM-RoBERTa

Additionally, we utilize a multi-language XLM-RoBERTa model, which is a variant of the RoBERTa model designed to handle multiple languages, including but not limited to English, Spanish, and other widely spoken languages. This model is particularly well-suited for multilingual tasks because it has been trained on a large-scale multilingual dataset, making it robust in understanding different linguistic structures and nuances across various languages. To enhance the model's capacity to capture complex representations, the XLM-RoBERTa model is augmented with two additional layers of neurons, which increase the model's depth, allowing it to learn more intricate patterns and relationships within the data. These layers are followed by GELU (Gaussian Error Linear Unit) activation functions, which are known for their smooth and non-linear transformations that help the model learn more efficiently and avoid issues like vanishing gradients, improving the overall performance of the model on the task at hand. This architecture ensures that the model is capable of handling both language-specific nuances and cross-linguistic patterns, providing a comprehensive solution for multi-language text classification tasks, such as the detection of sexist language in tweets. Both models used in this study are trained for three epochs, utilizing the Trainer API provided by the Hugging Face Transformers library. This approach is well-suited for the nature of our dataset, allowing for effective optimization of the model while minimizing the risk of overfitting.

Class	Precision	Recall	F1-Score	Support
Non-Sexist	0.79	0.55	0.65	1489
Sexist	0.65	0.85	0.74	1498
Accuracy			0.70	2987
F1 Macro			0.74	2987
Weighted Avg	0.72	0.70	0.69	2987

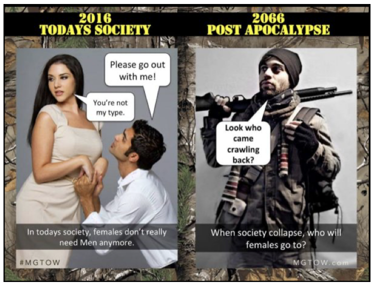
Table 4: Classification report of the XLM RoBERTa

Figure 3: Cross Entropy Loss over 3 epochs

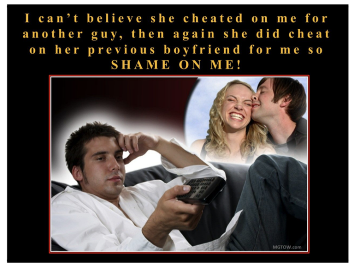
6 Memes Tweets

The dataset used for this study consists of 4044 tweets, which are balanced with respect to both language and labels. Of the tweets, 50.3% are in Spanish and 49.7% are in English. In terms of labels, 65.83% of the tweets are labeled as sexist, and the remaining 34.17% are labeled as non-sexist. These tweets, which may contain both text and images, have the text already provided, so no text extraction is required. To prepare the data for model training, we performed a stratified train-test split, ensuring that the distribution of labels (sexist vs. non-sexist) is consistent across both sets. The dataset is split into 80% for training and 20% for testing. The text data is tokenized using the pre-trained `XLMRobertaTokenizer`, and both the tokenized text and corresponding labels are used for training the model. The imbalance in the dataset, with a higher proportion of sexist tweets, is left unaddressed in the splitting process, but alternative techniques such as oversampling or undersampling could be considered during training to mitigate any potential bias.

We implement a multimodal system combines visual and textual processing for meme classification, using a ResNet-50 convolutional neural network to extract image features and XLM-RoBERTa for text understanding. The architecture processes images through standard ImageNet normalization and resizing, while text inputs are tokenized with a 256-length limit. The model fuses 2048-dimensional image features with 768-dimensional text embeddings (extracted from the [CLS] token) through concatenation, followed by a fully connected classification head. Implemented in PyTorch, the system employs an AdamW optimizer (lr=1e-5) and cross-entropy loss, training for 3 epochs with batch size 16. This approach effectively handles the dual-modality nature of memes, leveraging transfer learning from pretrained vision and language models while maintaining computational efficiency through attention masking and GPU acceleration. The modular design allows for straightforward adaptation to other multimodal classification tasks in social media analysis.



(a) English sexist Meme



(b) English Non sexist Meme

(c) Spanish sexist Meme

(d) English Non sexist Meme

Figure 4: Sample of Memes

Class	Precision	Recall	F1-Score	Support
Non-Sexist	0.4	0.49	0.44	276
Sexist	0.7	0.62	0.66	533
Accuracy			0.58	809
F1 Macro			0.66	809
Weighted Avg	0.6	0.58	0.58	809

Table 5: Classification report of the XLM-RoBERTa+CNN

Conclusion

Large transformer-based models deliver strong performance, as demonstrated by our implementation. However, their computational demands are significant, requiring substantial resources. In contrast, more concise methods like logistic regression and Naive Bayes can be highly effective—provided the data is meticulously cleaned and preprocessed. These approaches offer a compelling balance between efficiency and performance, especially when resource constraints are a consideration.

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